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Date:12/11/2022

Class: Cyber Forensic

Case File: Endangered Owl Trafficking Ring

- ❖ Illinois Institute of Technology, which is Investigating Agency called investigator Alaa Al Saad investigated about Endangered Owl Trafficking Ring to find evidence against suspect Sarah McAvoy and her partner to build a report with supporting evidence by examining digital images to prove the crime that commits by the suspect to providing to prosecuting attorney.

3)

a) Find the date and time of the owl delivery, if you can, and provide supporting evidence.

case about crime owl trafficking - Autopsy 4.19.3

Case View Tools Window Help

Listing Databases Table Thumbnail Summary

33 Results

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
wifi.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	16394	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Age
widgetpreviews.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	20480	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
user_dict.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	16394	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
timelydata.db			3	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	4096	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
textnow_data.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	139264	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
suggestions.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	20480	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
smhistory.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	20480	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
recents.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	36864	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
partnerbookmarks.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	16394	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
package_verification.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	32768	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
newsweather.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	65536	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
monitor.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	20480	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
mmssms.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	16394	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Age
ls.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	28672	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
localappstate.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	53248	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
local_trash.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	24576	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
library.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	28672	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
launcher.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	172032	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad
pr_gstore.db			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	32768	Allocated	Allocated	unknown	[LogicalFileSet1] LG Nexus 5 Image (Ad

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Table sms 21 entries Page 1 of 1 Export to CSV

id	body	address	type	date	date_sent	read	thread_id
1	Do you know Loggie Bear? Send them a friend request: https://fb.com/229775647Cw6IDw	32665	1	2017/02/06 18:30:55	2017/02/06 18:30:55	0	5
2	Sarah, you have 4 new notifications on Facebook: https://fb.com/1aau87KLV930x0M	32665	1	2017/02/06 18:53:19	2017/02/06 18:53:19	0	5
3	Your phone is activated! Reply HELP to learn more. Reply OFF to stop receiving notifications. Std msg...	40404	1	2017/02/01 11:00:00	2017/02/01 11:02:51	0	6
4	Your Twitter confirmation code is 636179.	40404	1	2017/02/01 11:00:00	2017/02/01 11:02:44	0	6
5	Thank you!	+13045184333	2	2017/01/31 18:40:00	2017/01/31 18:40:12	1	12
6	Sarah, the delivery is today 7 tonight the confirmation will come later through pidgin	+13045184333	1	2017/01/31 18:40:00	2017/01/31 18:40:12	1	12
7	Microsoft Verification Code: 1180	732873	1	2017/01/30 17:00:00	2017/01/30 17:01:49	0	7
8	Sarah, you have 1 new friend and 3 new notifications on Facebook: https://fb.com/10paz95FC7F1P	32665	1	2017/01/30 17:00:00	2017/01/30 17:01:49	0	5
9	Monica Neff commented on your post: https://fb.com/22d0gph6t1h472 Can't wait to hear about it...	32665103	1	2017/01/30 17:00:00	2017/01/30 17:01:47	0	11
10	Isaiah Dabner accepted your friend request. Reply to send him a message or visit his profile at https://fb.com/10paz95FC7F1P	32665102	1	2017/01/27 11:40:00	2017/01/27 11:40:55	0	10
11	Terry Bunch accepted your friend request. Reply to send him a message or visit his profile at https://fb.com/10paz95FC7F1P	32665101	1	2017/01/27 11:40:00	2017/01/27 11:40:48	0	9
12	Monica Neff commented on your post: https://fb.com/22d0gph6t1h472 Can't wait to hear about it...	32665103	1	2017/01/27 11:40:00	2017/01/27 11:40:45	0	8

Figure 1: screenshot shows the date and time of the owl delivery

Explanation: I found a message by the Autopsy program that shows an SMS message on raw under the SMS table inside mmssms.db, and the message is (the delivery is today 7 tonight, the confirmation will come later through pidgin) related to the time of delivery the owl. In addition, the raw show another information as time sends of message.

SQLite Carving Tool						
File Info						
data bases mmssms.db UNASSIGNED android_metadata						
Filter:						
	Offset	_id	body	address	type	date
0	003641	21	AT&T Free Msg: Learn about 3rd party mobile purchases & blocking at att.com/gophonemp	1111340061	1	20/01/20
1	0036B8	20	AT&T Free Msg: Your rate plan charge was successful. Your next rate plan charge will be on 02/18/...	28809090	1	20/01/20
2	00373F	19	AT&T Free Msg: Welcome to GoPhone! Refill, add packages, change plans & more at att.com/myg...	+11113271911	1	20/01/20
3	0037C7	18	66645 is your Facebook confirmation code #fb	32665	1	24/01/20
4	003810	17	Confirmed! To edit SMS preferences go to m.facebook.com/settings. To turn off all SMS messages	32665	1	24/01/20
5	0039C2	16	Your Twitter confirmation code is 600205	40404	1	25/01/20
6	003914	15	Use \$590 as Microsoft account security code	732873	1	25/01/20
7	00395D	14	Your Twitter confirmation code is 110497	40404	1	25/01/20
8	0039A3	13	Your phone is activated!Reply HELP to learn more.Reply OFF to stop receiving notifications.Std m...	40404	1	25/01/20
9	003A3E	12	Monica Neff accepted your friend request.Reply to send her a message or visit her profile at https...	32665100	1	27/01/20
10	003AB1	11	Terry Bunch accepted your friend request.Reply to send him a message or visit his profile at https...	32665101	1	27/01/20
11	003B84	10	Isaiah Dashner accepted your friend request.Reply to send him a message or visit his profile at htt...	32665102	1	27/01/20
12	003C2A	9	Monica Neff commented on your post.https://fb.com/l/2dDxpjpfetYcMa72 Can't wait to hear abou...	32665103	1	30/01/20
13	003CD9	8	Sarah, you have 1 new friend and 3 new notifications on Facebook: https://fb.com/l/8gaq9EkFC7F...	32665	1	30/01/20
14	003D59	7	Microsoft Verification Code: 1180	732873	1	30/01/20
15	003D98	6	Sarah, the delivery is today 7 tonight the confirmation will come later through pidgin	+13045184333	1	31/01/20
16	003E13	5	Thank you!	+13045184333	2	31/01/20
17	003E3C	4	Your Twitter confirmation code is 636179	40404	1	01/02/20
18	003E82	3	Your phone is activated!Reply HELP to learn more.Reply OFF to stop receiving notifications.Std m...	40404	1	01/02/20
19	003F1D	2	Sarah, you have 4 new notifications on Facebook: https://fb.com/l/alou8TVLWIOxcM	32665	1	06/02/20
20	003F8C	1	Do you know Loggie Bear? Send them a friend request: https://fb.com/l/23y7T6AoYCW9lDw	32665	1	06/02/20

Welcome
header is okay, seems to be an sqlite database file.
File format write version: 1 for legacy; 2 for WAL. 1
File format read version: 1 for legacy; 2 for WAL. 1
Bytes of unused "reserved" space at the end of each page. Usually 0. 0
Maximum embedded payload fraction. Must be 64.64
Minimum embedded payload fraction. Must be 32.64
Leaf payload fraction. Must be 32. 32

Figure 2: screenshot shows message delivery

Explanation: I found a message by the FQlite program that shows an SMS message on record when I clicked the SMS table inside mmssms.db, and the message is (the delivery is today 7 tonight. The confirmation will come later through pidgin) related to the time of delivery the owl. The picture was cut because the frame was bigger than the picture; therefore, I put half of the picture below.

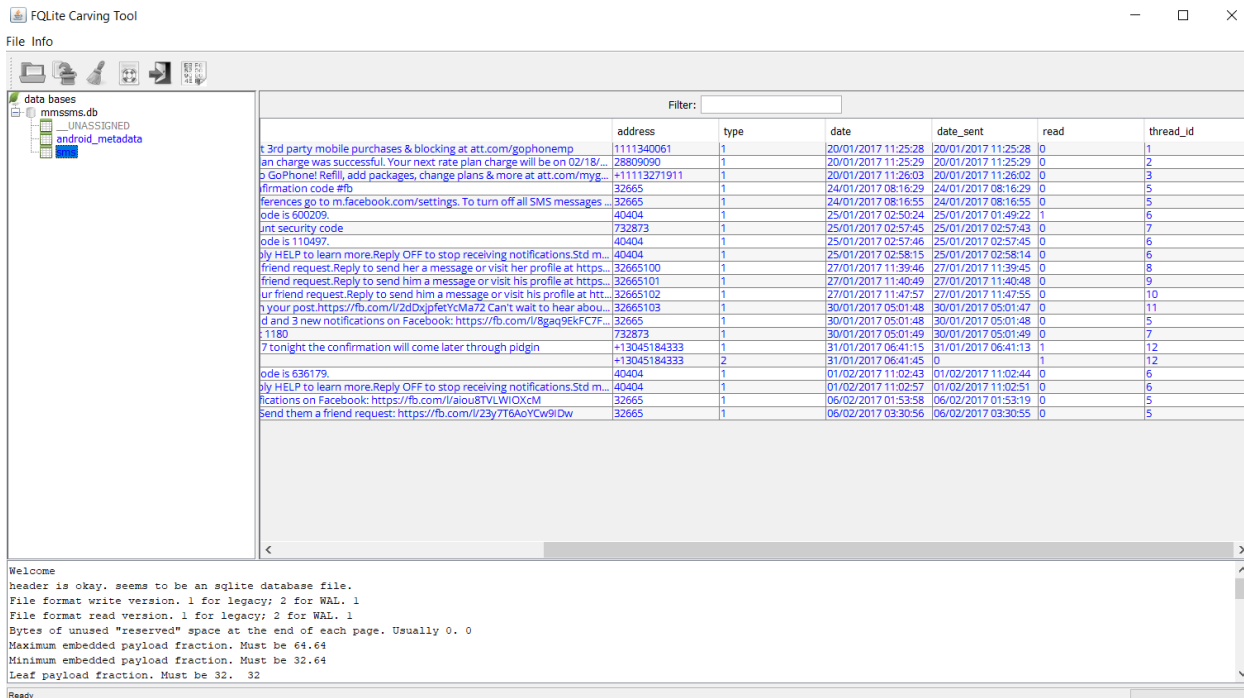


Figure 3 screenshot of the complete figure 2, which show the rest part of the screenshot

Explanation: the rest of the record shows other information such as (id, message body, address, type, date, data_send, read, thread_id) as time sends of message (31/01/2017 06:41:13).

b) Find out what identifying information you can find regarding this other party.

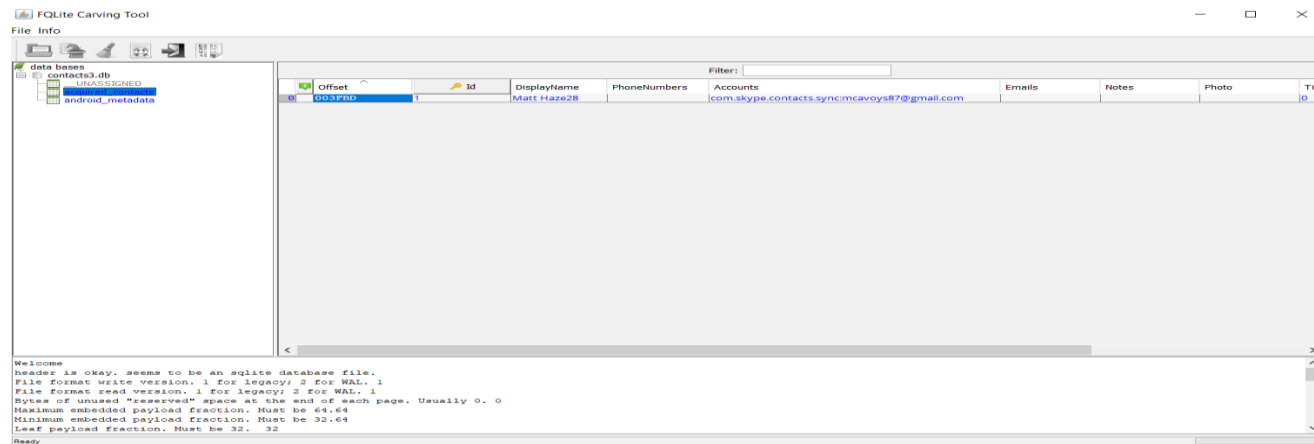


Figure 4: a screenshot shows the information the other party, who is the partner of Sarah McAvoy

Examination: I found the record on the FQLite program that shows the other party who is called Matt Haze28, and his account information when I clicked (acquired_contacts table) inside contacts3.db that I opened inside FQLite tool.

5) Write a report that summarizes how you conducted your investigation (aka investigation summary). Be sure to include:

Law enforcement has received information that two individuals were involved in owl trafficking. Therefore, the investigators obtained a search warrant for Sarah McAvoy, who is accused of committing illegal activity to seize her smartphone and her hard disk drive from her personal computer. I started to use two forensic tools; one is an Autopsy tool, and the other one is FQLite carving tool and google browser, which assist an investigator in successfully get acquisition the evidence that the law required to prove that Sarah McAvoy broke the law by committing owl trafficking ring and the other partner who deals with her.

An autopsy is a forensic tool that is a graphical interface to The Sleuth Kit and other digital forensics tools. This tool assists in focusing on forensic analysis of computer mass storage. Moreover, Autopsy is used by the military and law enforcement which helps them recover or find evidence on criminal devices to use against on trial; investigators can use various features on Autopsy to deal with data, such as timeline analysis for an Advanced graphical event viewing interface and a keyword search that is indexed in order to find files that mention relevant terms (Open Source Digital Forensics, n.d.). FQLite is a utility for locating and restoring deleted records and tables in SQLite databases. As a result, it searches the database for deleted records. These records and tables are recoverable and displayable (Pawlaszczyk, n.d.). These two tools assist me in finding digital evidence to build the case by collecting the evidence for law.

❖ The key steps I took in my investigation

Step one: I start to extract the folder of forensic copy to examine on the Autopsy program by selecting a new case from the splash screen window. Then I write the name of the case on the name case information window and click on next to go next step. I create a case by filling in the blanks with information (number of cases, name of Investigator, and note about the case), then press finish. On add data source window, I clicked next without any changes. In the next step, I select the type of data source logical files then next to add the location of the folder that I examine, then press next to configure ingest modules by selecting the feature I want to use and click next to open the forensic copy on autopsy then press finish on add data source window.

Step two: As Investigator, I display the information about the case that I fill by clicking the case tab. Then from the list, I select from case details.

Case Details

Case

Case Name:

case about crime owl trafficking

Case Number:

1001

Created Date:

2022/12/10 14:50:31 (CST)

Case Directory:

R:\student\alsaad\final\case about crime owl trafficking

Case Type:

Single-user case

Database Name:

R:\student\alsaad\final\case about crime owl trafficking\autopsy.db

Case UUID:

case_about_crime_owl_trafficking_20221210_145031

Examiner

Name:

Alaa Al Saad

Phone:

Email:

Notes:

2-sell motivation
3-text to schdual delivery

Organization

Name:

Not Specified

Point of Contact:

Phone:

Email:

Edit Details

Close

Figure 1: screenshot shows case details

Examination: I opened case details that I fill the information or applied when I started the case.

Key Step three: I started to expand by clicking file views, then I expanded file type under file views to also expand by extension to click on the image inside of it to find any evidence related to case in which found the suspect has an image about owl which shows proof that she interesting on it.

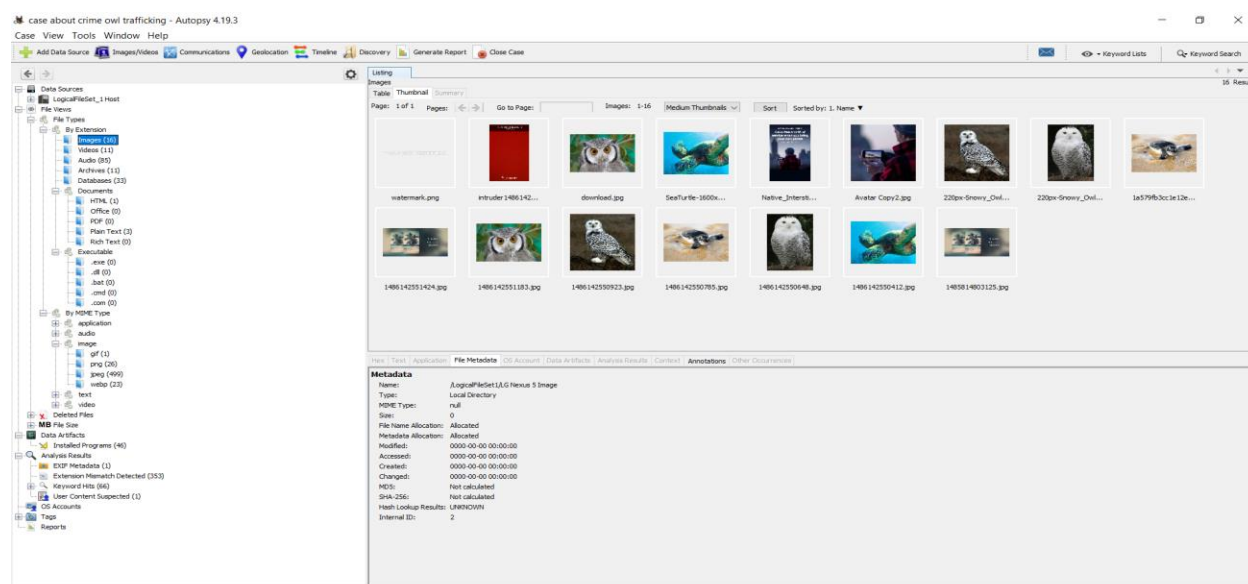


Figure 2: screenshot shows a different image.

Examination: after I click the image under the extension on the tree viewer, the picture shows on the result viewer, which I make display as an icon of the picture, not as a table, by clicking on a thumbnail on the result viewer.

Step four: I create a tag for every picture related to the case that shows the suspect's interest in the owl pictures by clicking on the picture right click then adding a file tag on the second list. I click new tag to pop up a window of create tag, which writes the name of the tag, then I check on the box to make the picture notable, then press ok.

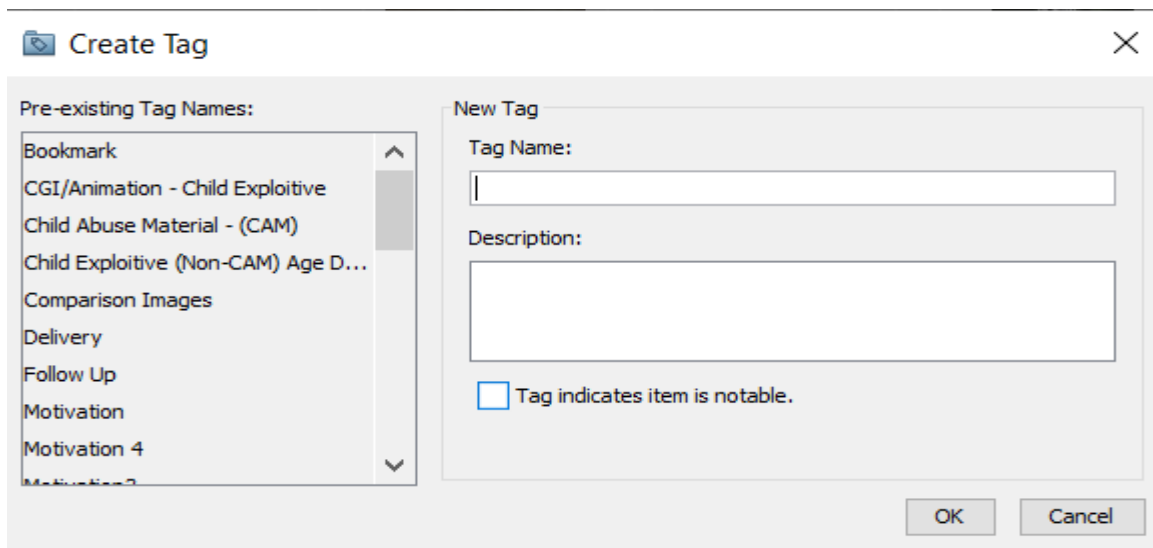


Figure 3: screenshot shows of creation tag.

Examination: I filled in the blank of the tag name "Owl interest" in every picture that Indicates the owl, then I made a notable red sign by clicking on the box. So, every picture that I tag will show under the tag area.

Step five: I extract the picture by right click on the picture and then extracting the file, which will take it to the folder extract under the folder of the name of the case I created.

Step six: I searched for other evidence by clicking on the tree area, but I did not find anything related to the case, such as video and audio, and others, so I completed the search until I clicked on a database which a number of databases around 33 databases. But I examine four databases (calendar.db, mmssms.db, contacts.db, wifi.db) on the result viewer to look for various table from the list on the content viewer to show records that have evidence against the suspect. Moreover, some tables were empty or did not have related information until I reached mmssms.db, which is message text.

Step seven: by selecting the table SMS, which is message text, I found records, and one of the records shows a message about the time and data delivery of owl, which makes me tag the database and then extract it.

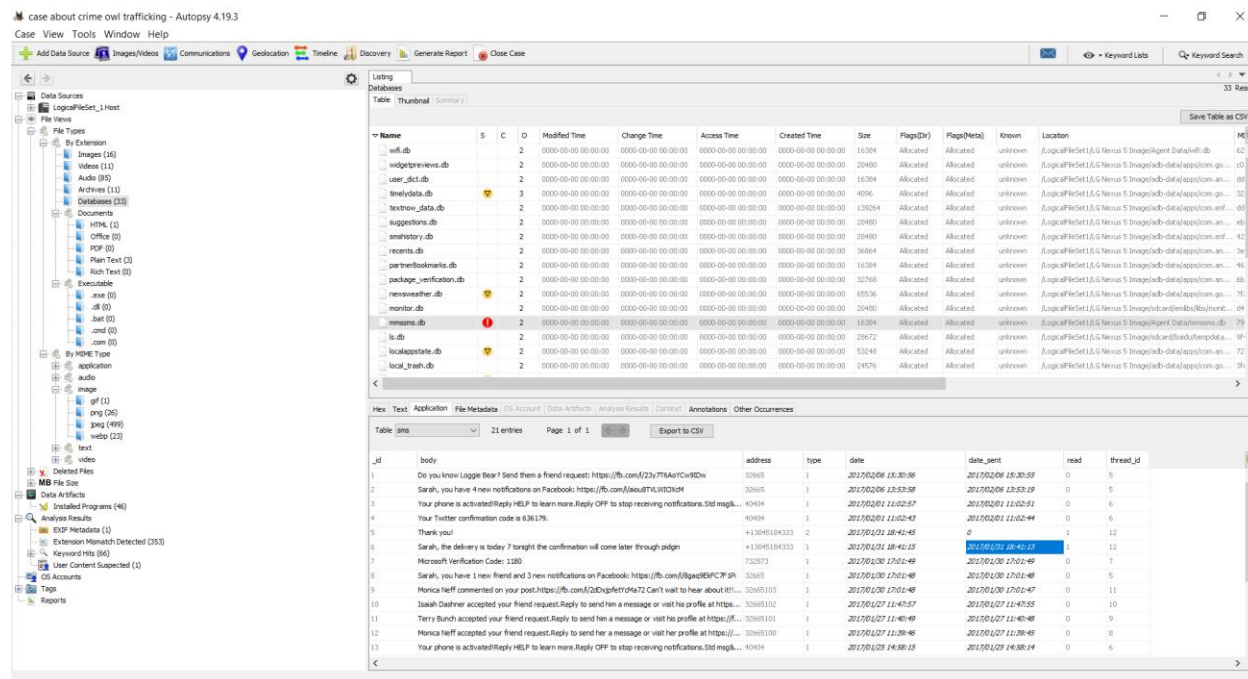


Figure 4: screenshot shows table SMS and its records of mmsms.db.

Examination: after I selected the database on the result viewer then, I clicked on the list table to examine, which assists me in seeing the message that proves the suspect broke the law with someone else. Therefore, I created a tag and called delivery for the database, and I made it notable after that I extracted as I did above.

Step eight: I search about other evidence by click on tree viewer, but I did not find anything relate to case until I got to image under MIME type which shows various image related to serial movies about harry potter and magicians.

Note: From my point of view, I selected all things related to the Harry Potter movie, from book to character to logo, that indicate to magicians because in the serial harry potter, every character has a pet, and the hero has an owl. Moreover, in the other scenes in the movie, people in the school of magic use owls to send and receive a message by owl. In addition, the number of pictures that indicate she is obsessed with owls and magicians leads me to think her motivation is to buy an owl to imagine herself as a magician. I know that sounds weird, but there are a lot of crimes in the world that do not have logical motivation rather than being obsessed with strange things. Another reason that made me tag and extracts every picture related to magicians is that I observed when I wrote on google browser magicians and owl. The browser showed pictures of an owl and stick that uses a wizard as a mascot. Moreover, I accounted for some pictures like the dark picture because of the feature of magicians and owls working in the dark or night.

Step nine: I create a tag for every picture related to the case that shows the suspect by clicking on the picture right click and then adding a file tag on the second list. I click the new tag to pop up the window of creating a tag, which writes the name of the tag, then I check on the box to make the picture notable, then press ok, then I extract it by clicking on picture extract the file.

Step ten: I did not find any information when I clicked on the communication, Geolocation, and discovery tabs on specialized viewers on the top toolbar. Moreover, I did not also find user activity recent file information on the data source summary.

Step eleven: I generated a report by clicking on generate report above the Result viewer, which will show a pop-up window where I need to choose HTML Report and fill header name and footer name, then press next and then select the data source, then press next. After that, I selected tagged result because I require to select items that had been tagged with notable tags. I made it to include in the report then I closed the load pop-up window. Finally, I can see the report by clicking on a report on the tree viewer.

The link of HTML Autopsy report is :

https://drive.google.com/file/d/1ClqDf54jtOO5wm99paSqRtH3g09Xahmr/view?usp=share_link

According to Former FBI Profiler Clint Van Zandt: "Motive is important because without an understanding of why people commit certain crimes in the way they do, we are left to begin at square one on every investigation, something we simply don't have the time or the resources to do (Zandt, 2006)."

I think I divided them into three categories because of the process investigators do when they are dealing with crime which it is working as a guideline to let us understand how investigators organize their work by searching about these categories. I watched documentary programs about criminals, investigators, lawyers, and judges, and every suspect has an interesting like something he or she searches about how to commit a crime before committing it to avoid catch by someone. In addition, criminals have reason to do crime even though there are no excuses for crime but understanding the motivation provides investigators experience with how people's brains work, which assists in Press accusations and solving the case quickly because their method of work is organized in terms of interesting, motivation, and finally the evidence to punish the criminals by the power of law.

I determined the date and time of the scheduled owl delivery on question three on begging of the document by examining the database of mmssms on autopsy and SQLite forensic tools of question 3, which is part of the explanation (look to question 3 answer and picture) that guide me to time and date by click on the time and date right click on date and time raw then display as the date which assists me to know date and time.

I identified information that the address of the other party is (+13045184333). In addition, I found the name of the person and his account, which identifies him a name: Matt Haze28, and his account: com.skype.contacts.sync:mcavoys87@gmail.com on contant3.db on autopsy and FQLite tools. Therefore, with this evidence, the police can catch him to give him to the court.